

Challenge (Habanero)

- Q1.** In a gaming scoreboard, a player's score is 4321. What is the value of the digit 4?
(A) Ones place
(B) Tens place
(C) Hundreds place
(D) Thousands place
(E) Ten thousands place
- Q2.** A coder used 658 MB of data. Rounded to the nearest ten, how much data was used?
(A) 650
(B) 660
(C) 700
(D) 600
(E) 500
- Q3.** In a coding logic problem, the number 945 is given. What is the value of the digit 9?
(A) Ones place
(B) Tens place
(C) Hundreds place
(D) Thousands place
(E) None of the above
- Q4.** A gamer's score is 1200. What is the expanded form of this number?
(A) $1000 + 200$
(B) $1000 + 200 + 0$
(C) $1000 + 20 + 0$
(D) $1000 + 2 + 0$
(E) $100 + 20 + 0$
- Q5.** A coding challenge requires a data usage of 753 GB. Rounded to the nearest hundred, how much data is required?
(A) 700
(B) 800
(C) 750
(D) 500
(E) 600
- Q6.** In a tech review, a smartphone's storage is 8192 MB. What is the standard form of this number?
(A) 8×10^3
(B) 8×10^2
(C) 8×10^1
(D) 8×10^4
(E) 8×10^0
- Q7.** A programmer used 279 hours of coding time. Rounded to the nearest ten, how many hours were used?
(A) 280
(B) 270
(C) 290
(D) 300
(E) 200
- Q8.** In a gaming tournament, a team's score is 4632. What is the value of the digit 6?
(A) Ones place
(B) Tens place
(C) Hundreds place
(D) Thousands place
(E) None of the above

Open Ended Questions (FRQ)

- Q9.** Write the number 9432 in expanded form. Show your work and explain your reasoning.
- Q10.** Round the number 6589 to the nearest hundred. Explain why you chose to round up or down.

Chef's Tips

- Tip 1: Emphasize the importance of place value in coding and data analysis.
- Tip 2: Encourage students to use real-world examples to illustrate their understanding of place value.
- Tip 3: Remind students to show their work and explain their reasoning in free-response questions.